

TRUNG VU

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Education

Ph.D. in Mathematics, Yale University

2021– Present

Advisor: Prof. Ivan Losev

B.Sc. in Mathematics, HKUST

2017–2021

Research Interests

Geometric representation theory, Quantum Algebra, quantum groups at roots of unity, Soergel bimodules and related objects.

Publications

- T. Vu, *On the functor relating Harish-Chandra bimodules and Soergel bimodules*. arXiv: 2507.17067
- I. Losev, A. Tsymbaliuk, T. Vu, *On De Concini-Kac forms of quantum groups*. arXiv: 2601.06696
- T. Vu, *Quantum Harish-Chandra bimodules at roots of unity and affine Hecke category*. arXiv: 2606.26331 (Slides)

Invited Talks

- Geometry, Physics, and Representation Theory seminar, Northeastern University *Dec 2026*
- AMS Fall Central Sectional, University of Minnesota, Duluth *Oct 2026*
- Geometric Representation Theory seminar, Yale *Sep 2026*
- Pure math seminar, HKUST (online) *Jul 2026*
- Algebraic and Geometric Representation Theory, University of Virginia (short talk) *Jul 2026*
- Representation and Lie Theory seminar, University of California, Riverside *May 2026*
- AMS Eastern Sectional Meeting, Boston College, [slides](#) *Mar 2026*
Quantum Harish-Chandra bimodules at roots of unity
- Seattle Noncommutative Algebra Conference, University of Washington, [slides](#) *Dec 2025*
On De Concini-Kac forms of quantum groups
- Physics and Math seminar, Purdue University *Mar 2025*

Contributed Presentations

- Categorification and Symplectic Duality Workshop, Northeastern University (Poster) *Jun 2025*

Teaching

Certificate of College Mathematics Teaching Preparation, Yale University *2027*

Instructor of Record

- Math 1200 Calculus of Functions of Several Variables, Yale University *Spring 2027*
- Math 1150 Calculus of Functions of One Variable II, Yale University *Fall 2025*
- Math 115 Calculus of Functions of One Variable II, Yale University *Spring 2025*

Teaching Assistant

- Math 115 Calculus of Functions of One Variable II, Yale University *Spring 2024*
- Linear algebra, Yale University *Spring 2023*
- Math 112 Calculus of Functions of One Variable I, Yale University *Spring 2022*

Grader

- Calculus, Linear algebra, Abstract algebra, Introduction to differential manifold.

Mentoring

- SUMRY *Summer Undergraduate Math Research at Yale* (poster) *Summer 2026*

Workshops and Conferences Attended

- Representation Theory Under the Influence of Quantum Field Theory and Motivic Homotopy Theory, SL Math *Aug 2026*
- WARTHOG 2025, University of Oregon, *Cluster Algebras and Braided Varieties* *Aug 2025*
- 7-th Canada-Mexico-US meeting, University of Southern California, *Representation Theory, Noncommutative Algebra, and Categorification* *Jul 2025*
- Advances in Representation Theory, Northeastern University *Jun 2025*
- Quantum Groups and Representation Theory, North Carolina State University, *On the occasion of Kailash Misra's 70 birthday* *Oct 2024*
- WARTHOG 2024, University of Oregon, *Coherent-constructible equivalences in local Geometric Langlands and Representation Theory* *Jul 2024*
- MSRI summer school, *Koszul Duality in the Local Langlands Program* *Jul 2024*
- The 14th Southern Lie Theory Workshop series, University of Virginia, *Quantum structures on Lie Theory* *Mar 2024*
- Summer school in Geometric Representation Theory, MIT, *Coulomb Branches and Knot Homology* *Jun 2023*